



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : Electricity Bill Management System

Group No : 06, CSE370 Lab Section : 06, Spring 2026		
ID	Name	Contribution
23201376	Mir Mohammad Sajedul Islam	Frontend+Backend
23201378	Bishal Golder	Frontend+Backend
23201031	Salman Naguib	Frontend+Backend

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Introduction

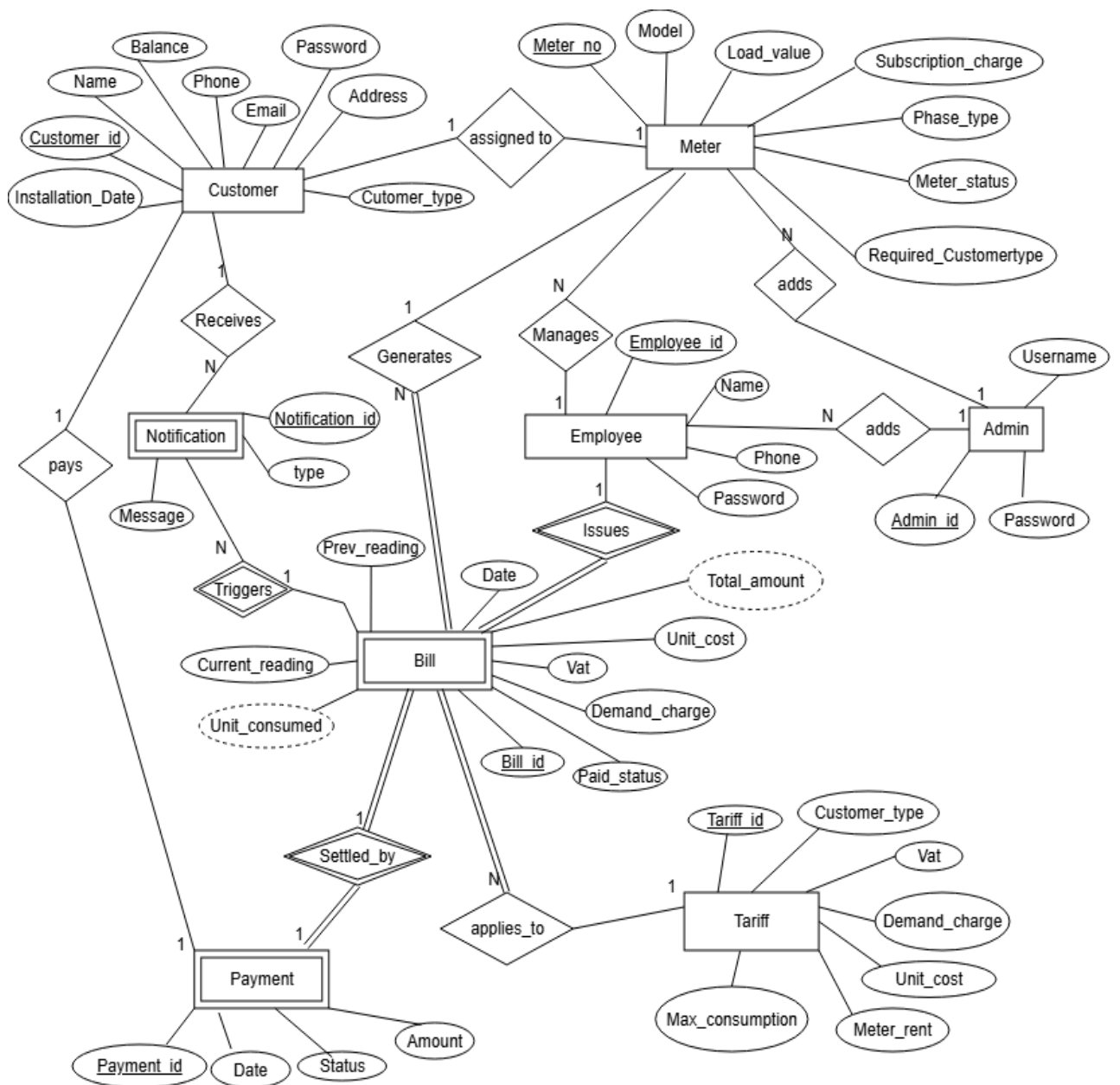
SparkEnergies is a web-based platform created to replace the outdated electricity billing and distribution system in Bangladesh. The project was born out of a need to tackle some major challenges in the current setup—like manual meter readings, paper-based tracking, and the frequent billing errors that frustrate both consumers and utility providers. We built SparkEnergies with one clear goal in mind: to make the whole process of electricity billing smoother, more transparent, and efficient. By automating everything from meter readings to tariff calculations, the system eliminates human errors and makes the process faster and more reliable. So, there will be no more waiting for paper bills or worrying about "ghost billing" discrepancies.

The platform covers all aspects of electricity billing, whether it's for residential customers or large industrial users. It uses a centralized, automated system that applies the right tariffs for each consumer sector, and a 28-day algorithm ensures that duplicate billing won't happen. Consumers get a clear, easy-to-use digital portal where they can track their energy usage and pay bills instantly. Through this project, we aimed to gain hands-on experience in web application development, database design, and system-oriented problem solving. On the operational side, SparkEnergies also makes life easier for utility workers by automating route assignments and reducing manual tasks. In a country where modernization is key, this system is part of Bangladesh's vision for a smart, sustainable, and paperless future.

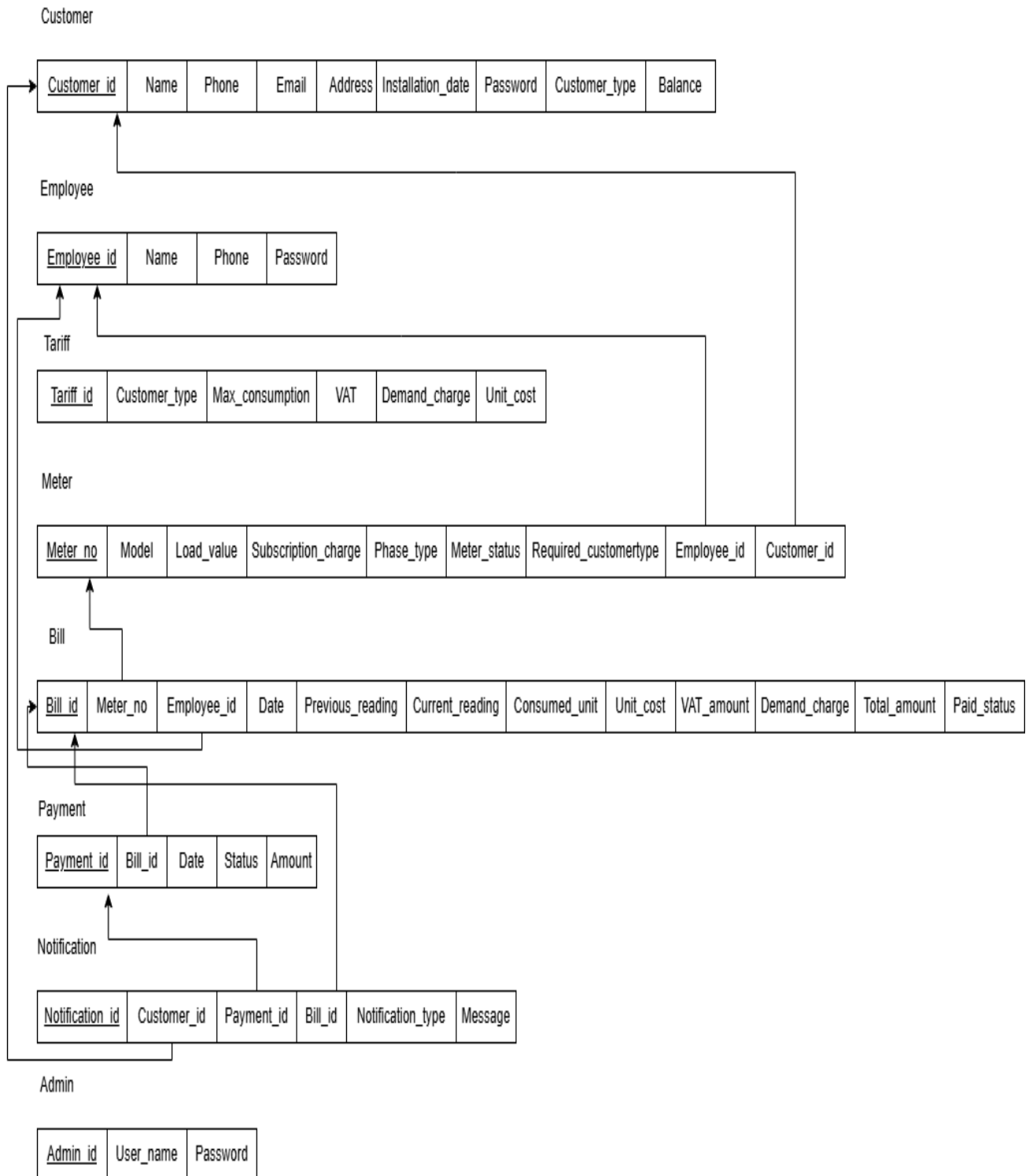
Project Features

ID, Name	Features [3 per member]	
23201376, Mir Mohammad Sajedul Islam	Ft 1	Digital Bill Payment: View detailed billing history, distinguish between paid/unpaid statuses, and pay pending electricity bills securely using the account's wallet balance.
	Ft 2	Meter Application & Setup: Browse matching meter blueprints based on user category, auto-deduct the subscription fee via digital wallet, and instantly receive an assigned physical meter.
	Ft 3	Visual Analytics & Invoicing: Track 6-month electricity usage trends through a dynamic interactive graph and download auto-generated, formatted PDF invoices for personal records.
23201378, Bishal Golder	Ft 1	Assigned meter Tracking and Billing History: View a personalized list of assigned meters, track physical addresses for field visits, and monitor the last recorded reading dates to prioritize pending billing. Also, bill creation history can be viewed for a particular employee.
	Ft 2	Dynamic Bill Generation: Select an assigned meter, input current readings, and auto-calculate the total bill (VAT, Demand Charge) based on specific tariff slabs.
	Ft 3	Data Validation & Safety Lock: Enforce strict data integrity by blocking illogical meter inputs and utilize a 28-day lock to prevent premature or duplicate bill generation.
23201031, Salman Naguib	Ft 1	Wallet & Employee Management through Admin: Search for specific customers to securely recharge their account balance and register new field employees into the system.
	Ft 2	Unpaid Bill queue & Customer Directory Viewing: Cross-references multiple tables to highlight who is due on bills and the customer directory, which lists identifying information and balances for registered customers.
	Ft 3	Inventory & Categorization: Add new meters into the global inventory, and each meter is automatically assigned an 'available' or 'deployed' status upon entry based on the deployment status.

ER/EER Diagram



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.

Yes, our converted schema is already in 1st normal form because there is no multivalued or composite attribute in the relations. Also, there are no nested relations.

- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?

Yes, our converted schema diagram is already in 2NF. Because it is already in 1NF and there is no partial functional dependency. There can not be any partial functional dependency in our relational schema because all relations in our schema have only one primary key. So, all non prime attributes are functionally dependent on one primary key.

- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

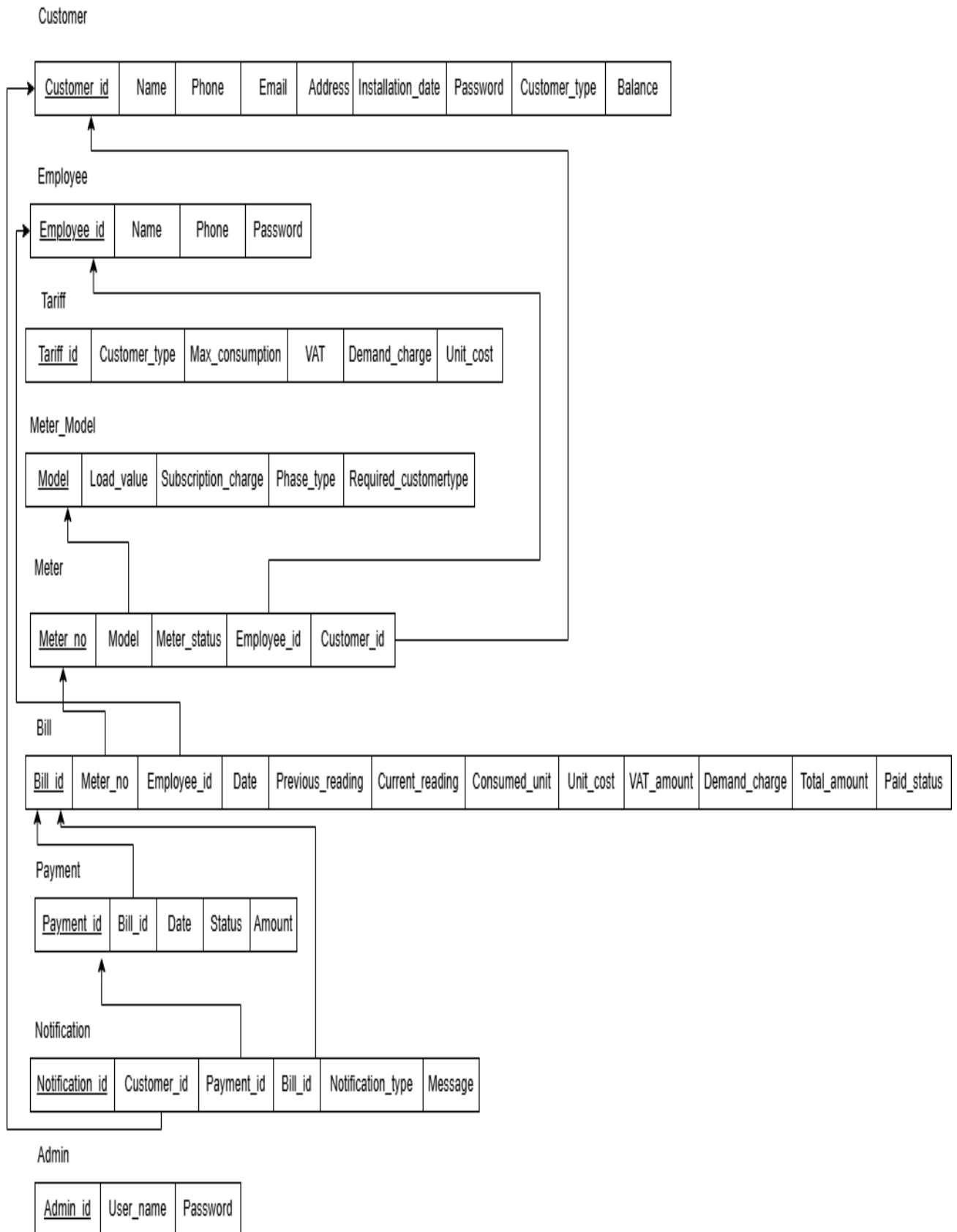
No, our converted schema is not in 3NF. Because in the relation named “Meter” in our schema, Load_Value, Subscription_charge, Phase_type and Required_customertype attributes are functionally dependent on Model and Model is functionally dependent on Meter_no. So, there was transitive dependency.

Meter_no → Model, Employee_id, Customer_id

**Model → Load_Value, Subscription_charge, Phase_type,
Required_customertype**

To convert this schema to 3NF, the Meter relation needs to be splitted to Meter_model and Meter. After that, there would not be any transitive dependency.

The 3rd Normal Form of the schema diagram is below-



Frontend Development

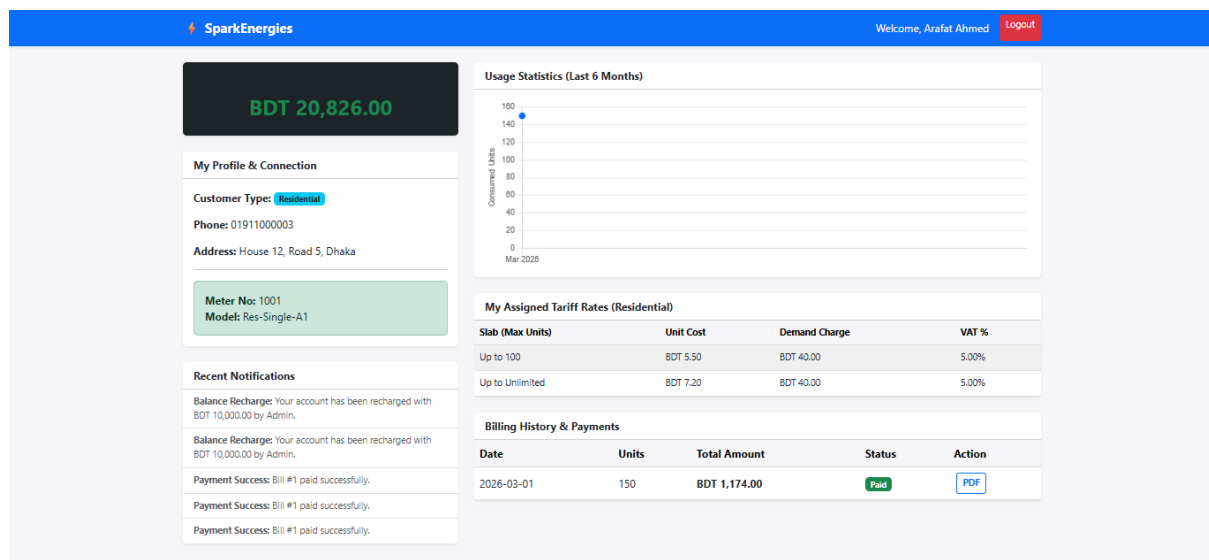
Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 23201376, Name : MIR MOHAMMAD SAJEDUL ISLAM

Developed the Customer Dashboard UI to create an intuitive, secure, and mobile-responsive self-service portal for electricity consumers using HTML, CSS, and Bootstrap 5. The interface is designed to simplify financial transactions, prevent application errors, and make account management effortless.

Digital Bill Payment:

- Designed a structured layout utilizing Bootstrap cards to clearly separate the digital wallet balance display from the active billing ledger.
- Implemented a responsive history table using table-hover and align-middle classes, ensuring billing months and financial amounts are highly readable across all devices.
- Utilized Bootstrap contextual badges (bg-success for Paid, bg-danger for Unpaid) to provide instant visual distinction of invoice statuses at a single glance.
- Integrated prominent, state-aware action buttons (btn-primary for Pay Now, and disabled btn-outline-secondary for settled bills) to securely guide the user's payment workflow without confusion.



The image shows a mobile application interface for SparkEnergies. At the top is a blue header with the SparkEnergies logo. Below the header is a dark grey card displaying the balance 'BDT 20,826.00' in green. Underneath is a white card titled 'My Profile & Connection'. This card contains the following information: 'Customer Type: Residential' (with 'Residential' in a blue pill), 'Phone: 01911000003', and 'Address: House 12, Road 5, Dhaka'. At the bottom of the white card is a light green pill-shaped box containing 'Meter No: 1001' and 'Model: Res-Single-A1'.

Meter Application & Setup:

- Developed a clean application interface housed within a dedicated Bootstrap card to keep the setup process focused and uncluttered.
- Designed dynamic, category-specific display logic that only presents the user with meter blueprints compatible with their assigned customer category, preventing hardware mismatch errors.
- Integrated interactive visual feedback for the submission process, changing the Apply button to a disabled Processing state utilizing Bootstrap's spinner-border component to prevent accidental double-clicks and duplicate subscription deductions.

My Profile & Connection

Customer Type: Residential

Phone: 0171100004

Address: Block D, Bashundhara R/A, Dhaka

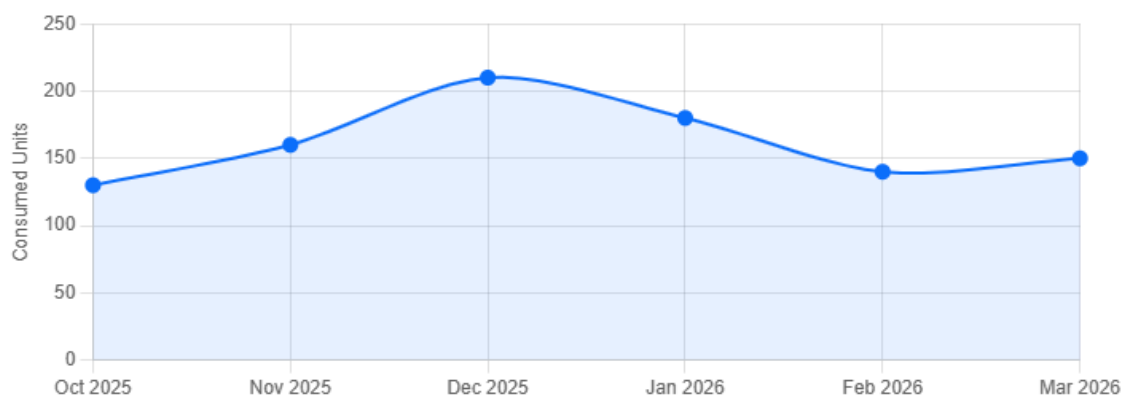
You do not have a meter assigned yet.

[Apply for Residential Meter](#)

Visual Analytics & Invoicing:

- Integrated a dedicated canvas container to render the 6-month interactive Chart.js graph, applying custom CSS transitions to the parent card to provide a premium, modern feel on mouse hover.
- Structured the layout to prioritize the analytics graph at the top of the dashboard for immediate visual impact upon login.
- Designed a clear, accessible Download PDF Invoice action button, supported by custom print CSS rules to ensure the generated invoice automatically hides web navigation elements and formats perfectly as a professional corporate document.

Usage Statistics (Last 6 Months)



BILLED TO:

Arafat Ahmed

House 12, Road 5, Dhaka

Phone: 01911000003

Type: **Residential**

CONNECTION DETAILS:

Meter No: 1001

Model: Res-Single-A1

Status: PAID

Description	Reading / Units	Rate (BDT)	Amount (BDT)
Previous Meter Reading	0	-	-
Current Meter Reading	150	-	-
Total Units Consumed	150	7.20	1,080.00
Demand Charge			40.00
VAT Amount			54.00
TOTAL AMOUNT PAYABLE			BDT 1,174.00

If you have any questions about this invoice, please contact SparkEnergies Support.

Thank you for your business!

Contribution of ID : 23201378, Name : BISHAL GOLDER

Developed the Employee Dashboard UI to create a responsive workspace for field agents/employees using HTML, CSS, and Bootstrap 5. The interface is designed to minimize data entry errors and maximize route efficiency.

Assigned Meter Tracking and Billing History:

- Designed a clean, view-only data grid utilizing Bootstrap's table-striped and align-middle classes to display the employee's personalized route.
- Ensured physical addresses, customer names, and meter models are clearly visible for easy field navigation.

- Implemented visual status indicators (e.g., highlighting "Never Billed" using badge bg-warning) so employees can instantly prioritize which meters need to be checked first based on their last recorded reading date. At the same time, employees can view how many days have passed since the last reading date and if 28 days pass after reading a meter from the last time, that meter is shown in red color.
- Developed the "Recently Generated Bills" history table with visual status indicators, utilizing Bootstrap contextual badges (bg-success for Paid, bg-danger for Unpaid) so employees can instantly verify the financial status of their recent field operations at a single glance.

My Assigned Route (Meters)			
Meter No	Customer Name	Address	Last Reading Date
1001 Res-Single-A1	Arafat Ahmed	House 12, Road 5, Dhaka	May 1, 2026 (21 days ago)
1003 Res-Single-A1	Shafik Alam	Mohammadpur,Dhaka	May 1, 2026 (21 days ago)
1005 Const-Three-X1	MegaBuild Construction	10, Sadarghat, Dhaka	Never Billed
1007 Hosp-Three-H1	LifeCare Hospital	15, Mohammadpur, Dhaka	Apr 1, 2026 (51 days ago)
1008 Res-Single-A1	Rina Sultana	12/2, Banasree, Dhaka	May 1, 2026 (21 days ago)
1015 Res-Single-A1	Shilpi Islam	23, Rajshahi, Dhaka	Never Billed

Recently Generated Bills (By You)					
Bill ID	Date	Meter	Units Consumed	Total Amount	Status
#3	May 1, 2026	1003	40	BDT 281.00	Unpaid
#5	May 1, 2026	1001	144	BDT 1,138.64	Paid
#6	May 1, 2026	1008	60	BDT 396.50	Unpaid
#2	Apr 30, 2026	1004	200	BDT 3,015.00	Paid
#4	Apr 1, 2026	1007	500	BDT 3,975.00	Unpaid
#1	Mar 1, 2026	1001	150	BDT 1,174.00	Paid

Dynamic Bill Generation:

- Developed the bill generation interface using a structured card layout.
- Implemented a dropdown menu that only populates with the specific meters assigned to the logged-in employee, preventing them from accidentally billing out-of-route customers.
- Designed a streamlined numeric input field for the "Current Reading," keeping the UI uncluttered so the backend can seamlessly auto-calculate the VAT and Demand Charges upon submission.

SparkEnergies (Staff)

Bill successfully generated for Meter 1008! Total: BDT 396.50

Rahim Uddin

 01711000001

Generate New Bill

Select Assigned Meter

Meter 1008 (Rina Sultana) 

Current Meter Reading (Units)

60

System will auto-fetch previous reading.

Generate & Send Bill

Data Validation & Safety Lock:

- Integrated immediate visual feedback mechanisms into the UI to enforce system safety locks.

-  SparkEnergies (Staff)

Agent: Rahim Uddin

Logout

Current reading (200) cannot be less than previous reading (500).

Rahim Uddin

01711000001

Generate New Bill

Select Assigned Meter

Meter 1007 (LifeCare Hospital)

Current Meter Reading (Units)

200

System will auto-fetch previous reading.

Generate & Send Bill

My Assigned Route (Meters)

Meter No	Customer Name	Address	Last Reading Date
1001 Res-Single-A1	Arafat Ahmed	House 12, Road 5, Dhaka	Mar 1, 2026
1003 Res-Single-A1	Shafik Alam	Mohammadpur,Dhaka	May 1, 2026
1007 Hosp-Three-H1	LifeCare Hospital	15, Mohammadpur, Dhaka	Apr 1, 2026
1008 Res-Single-A1	Rina Sultana	12/2, Banasree, Dhaka	Never Billed

Recently Generated Bills (By You)

Bill ID	Date	Meter	Units Consumed	Total Amount	Status
#3	May 1, 2026	1003	40	BDT 281.00	Unpaid
#2	Apr 30, 2026	1004	200	BDT 3,015.00	Paid
#4	Apr 1, 2026	1007	500	BDT 3,975.00	Unpaid

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Contribution of ID: 23201031, Name: SALMAN NAGUIB

Developed the Admin Dashboard UI to create a streamlined, responsive, user-friendly workspace for the admin using HTML, CSS, and Bootstrap 5. The interface is designed to minimize data entry errors and maximize route efficiency. The dashboard is designed to serve as the central nervous system for the utility platform, enabling high-level oversight and management of infrastructure, personnel, and financial health.

Wallet & Employee Management through Admin:

- Designed a structured card-based layout to group critical administrative functions into clearly separated action panels.
- Implemented three primary control modules: Customer balance recharge, Employee registration, and Meter inventory addition
- Utilized Bootstrap's grid system to create a balanced three-column layout, ensuring equal visual importance and accessibility.
- Integrated full-width action buttons to ensure clear call-to-action visibility and prevent misclicks.
- Designed a tab-based navigation system to logically separate functionalities without page reloads.

The screenshot displays the SparkEnergies (HQ) Admin Dashboard. At the top, there's a header with the logo, system admin status (admin), and a logout button. Below the header is a navigation bar with tabs: Quick Actions, Customers & Bills, Meter Inventory, and Employee Directory. The main content area features three primary action panels:

- Recharge Customer Account:** Includes input fields for Customer ID and Amount (BDT), and a Process Recharge button.
- Register New Employee:** Includes input fields for Full Name, Phone Number (Login ID), and Temporary Password, and a Create Account button.
- Add Meter to Inventory:** Includes input fields for New Meter Number (with an example e.g. 1005) and Meter Model (with a dropdown menu for Select Blueprint...), and a Stock Meter button.

- Developed a clean and minimal staff management interface focused on clarity and quick reference.
- Used typographic emphasis (fw-bold, text-primary) to highlight key identifiers.
- Maintained a simple table layout to ensure fast scanning and usability without unnecessary complexity.

Active Staff Members		
Employee ID	Full Name	Phone (Login ID)
EMP-002	Karim Mia	01811000002
EMP-001	Rahim Uddin	01711000001

Unpaid Bill queue & Customer Directory Viewing:

- Developed a dual-section interface combining financial monitoring and customer records for efficient administrative oversight.
- Implemented a responsive table using table-hover and align-middle for enhanced readability.
- Highlighted outstanding balances using red typography and badges (bg-danger) to draw immediate attention.
- Structured chronological sorting (oldest first) to prioritize urgent payments.
- Designed a structured data table displaying customer identity, contact details, and account status.
- Used contextual badges (bg-info) to visually represent customer categories.
- Highlighted wallet balances in green text to improve financial readability.

⚡ SparkEnergies (HQ)

System Admin: **admin** [Logout](#)

🔧 Quick Actions

👥 Customers & Bills

📊 Meter Inventory

👤 Employee Directory

Unpaid Bills Queue

Bill ID	Date	Customer	Meter No	Amount Due	Status
#4	May 3, 2026	TechSoft IT (ID: 2)	1002	BDT 17,212.00	Pending Payment
#5	May 3, 2026	Arafat Ahmed (ID: 1)	1001	BDT 7,398.32	Pending Payment

Registered Customers Directory

ID	Name	Phone	Email	Type	Wallet Balance
6	Bishal Golder	32456865434	bishal@email.com	Residential	BDT 20,330.00
5	Donald J trump	01223456	donald@email.com	Factory	BDT 91,985.00
3	Green Farms	01511000005	farm@email.com	Agriculture	BDT 500.00
2	TechSoft IT	01311000004	info@techsoft.com	Corporate	BDT 15,000.00
1	Arafat Ahmed	01911000003	araf@email.com	Residential	BDT 20,826.00

Inventory & Categorization:

- Created a centralized inventory tracking interface for all system meters.
- Designed a comprehensive table displaying Meter number, model, phase type, target customer category, and assignment status (customer and employee linkage).
- Implemented status-based visual indicators by combining server-side logic (PHP) with frontend styling (Bootstrap 5) to dynamically reflect each meter's state in the user interface. This approach ensures that the status is both data-driven and visually intuitive. (Green badge → Available stock, Grey badge → Deployed meters).
- Structured sorting logic to prioritize available meters at the top(topmost section under deployed), improving operational efficiency in allocation tasks.


SparkEnergies (HQ)

System Admin: **admin**
Logout

 Quick Actions
 Customers & Bills

Meter Inventory

 Employee Directory

Global Meter Database

Meter No	Model Blueprint	Phase	Target Customer	Assigned Customer ID	Assigned Employee ID	Status
1004	Fac-Three-F1	Three Phase	Factory	5	2	Deployed (Assigned)
1003	Res-Single-A1	Single Phase	Residential	6	1	Deployed (Assigned)
1002	Corp-Three-C1	Three Phase	Corporate	2	2	Deployed (Assigned)
1001	Res-Single-A1	Single Phase	Residential	1	1	Deployed (Assigned)
1009	Agri-Single-G1	Single Phase	Agriculture	-	-	In Stock (Available)
1007	Hosp-Three-H1	Three Phase	Hospital	-	-	In Stock (Available)
1006	Edu-Three-E1	Three Phase	Educational_institute	-	-	In Stock (Available)
1005	Const-Three-X1	Three Phase	Construction	-	-	In Stock (Available)

Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 23201376, Name : MIR MOHAMMAD SAJEDUL ISLAM

Digital Bill Payment (PHP + SQL)

1. Implemented the backend logic to retrieve the logged-in customer's complete billing history, applying SQL filters to accurately categorize and render invoices based on their 'Paid' or 'Unpaid' status.
2. Developed a secure payment verification flow that checks the customer's digital wallet balance against the pending bill amount, blocking the transaction if there are insufficient funds.
3. Integrated synchronized SQL updates to process the payment, simultaneously deducting the billed amount from the user's wallet balance and updating the specific bill's database status to 'Paid' to ensure strict financial integrity.

Meter Application & Setup (PHP + SQL)

1. Created a backend filtering system that queries the database to display only the specific meter blueprints compatible with the logged-in user's designated customer category.
2. Implemented automated financial backend checks to verify the user's wallet balance and securely deduct the required one-time subscription fee upon application submission.
3. Engineered the automated assignment logic using SQL updates to instantly link an available physical meter from the inventory to the user's account, automatically changing the hardware's database status to 'Assigned'.

Visual Analytics & Invoicing (PHP + SQL)

1. Developed complex SQL aggregation queries utilizing date grouping and sum functions to accurately extract the user's electricity consumption data over the trailing 6-month period.
2. Ensured the analytics data fetch is strictly bound to the logged-in customer's ID via backend session validation, protecting data privacy and preventing cross-account data exposure.
3. Implemented backend data extraction and formatting logic to seamlessly pull specific invoice records from the database and feed them into the automated PDF generation engine for user downloads.

Contribution of ID : 23201378, Name : BISHAL GOLDER

Assigned Meter Tracking and Billing History (PHP + SQL)

1. Implemented the backend logic to fetch assigned meters from the database and render them dynamically on the employee's route dashboard.
2. Linked the route-fetching process to the logged-in employee's ID, ensuring field workers can only access and view their own designated customers.
3. Added database queries to retrieve and display the most recent billing date for each assigned meter, handling newly assigned meters so no data is dropped from the list.

Dynamic Bill Generation (PHP + SQL)

1. Implemented the bill generation backend to process submitted meter readings and securely store the newly generated bill in the database.
2. Developed a dynamic tariff calculation process that automatically fetches the correct unit cost, VAT percentage, and demand charge directly from the database based on the specific customer's category (e.g., Residential, Hospital).
3. Integrated the billing process with SQL inserts so that when a reading is submitted, the total amount is calculated accurately on the backend and saved with an 'Unpaid' status.

Data Validation & Safety Lock (PHP + SQL)

1. Enforced strict data integrity by adding backend checks to fetch and compare the submitted current reading against the last recorded reading in the database.
2. Added backend validation to block illogical inputs, automatically rejecting any new reading that is mathematically lower than the previous one to prevent negative billing.

3. Developed the 28-day safety lock logic, which checks the exact date of the last generated bill and strictly prevents the system from accepting a new bill if 28 days have not passed, ensuring no duplicate bills are created and the interval between bills remains at least a month.
4. Created a data fetch to retrieve the last 15 bills specifically generated by the logged-in employee, ordering them chronologically to provide an accurate, real-time audit data of their billing activities.

Contribution of ID : 23201031, Name : SALMAN NAGUIB

Inventory & Workforce Management (PHP + SQL):

1. Automated Staff Onboarding: Engineered a registration backend that processes new employee accounts, utilizing prepared statements to securely store credentials while preventing duplicate phone number entries via database exception handling.
2. Hardware Asset Management: Implemented an inventory logic that allows for the addition of new hardware by linking specific “Meter_no” entries to predefined “meter_model” blueprints.
3. Status-Aware Deployment Tracking: Developed a global meter database view using “join” operations to display the status of every unit in the inventory, tracking whether a meter is 'Available' or 'Assigned' and identifying the specific customer and employee linked to the hardware.

Financial Oversight & Revenue Queue (PHP + SQL):

1. Real-Time Debt Monitoring: Created a centralized Unpaid Bills Queue utilizing SQL filtering to aggregate all invoices with an 'Unpaid' status across the entire customer base.
2. Cross-Table Revenue Logic: Developed queries that link billing data with customer and meter records, enabling administrators to see exactly which users owe outstanding amounts and the specific meters associated with those debts.
3. Dynamic Blueprint Fetching: Integrated a backend system that pulls available meter models from the ‘meter_model’ table to populate administrative forms, ensuring that only valid, existing hardware blueprints can be added to the stock.

Automated Notification System (PHP + SQL)

1. Transactional Trigger Logic: Developed backend logic to automatically generate system notifications when specific financial or administrative actions occur, such as meter assignments or balance recharges.
2. Administrative Reconciliation Alerts: Integrated INSERT queries within the balance recharge flow to inform customers whenever an administrator manually adds BDT funds to their account.
3. Deployment Confirmation: Implemented automated messaging that triggers upon the successful assignment of a meter, providing the customer with their new Meter Number and account status update.
4. Relational Data Mapping: Engineered the notification backend to store unique Customer_id and Notification_type identifiers, ensuring alerts are accurately routed to the correct user dashboard.

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

https://drive.google.com/drive/folders/1XepPmhTQuaMx5qiJu4xYZB0MJPzEfACB?usp=drive_link

Conclusion

SparkEnergies successfully modernizes Bangladesh's electricity billing system by replacing manual, error-prone tracking with a fully automated web application. By integrating a PHP and SQL backend with a responsive Bootstrap 5 frontend, our team created dedicated portals for consumers, field employees, and system administrators. The platform seamlessly handles complex operations like dynamic tariff calculations, digital wallet payments, and real-time inventory management. Also, backend safeguards like the 28-day safety lock effectively eliminate the risk of duplicate or "ghost" billing, ensuring complete financial integrity.

From an academic standpoint, the development process offered valuable opportunities to apply theoretical knowledge within a practical context. Constructing the initial ER and EER models, and subsequently normalizing the relational schema to the Third Normal Form to eliminate transitive dependencies, reinforced the importance of disciplined database design in supporting scalable, reliable applications. Seeing those foundational concepts function effectively within a live system — managing real-time inventory and financial data — was a meaningful confirmation of their practical relevance.

Ultimately, SparkEnergies proves the value of a centralized platform by maximizing route efficiency for workers while giving consumers transparent, interactive tools to track their energy usage. This project stands as a practical and scalable step toward a smarter, paperless future for the nation's utility sector.

References

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2. Ken Swartwout. (2026, December 12). Connect to MySQL with PHP in XAMPP. YouTube <https://youtu.be/ueWpNe0PG34?si=xT6OcZkhFbqSmu27>
3. CodeWithHarry. (2020, February 11) . Php Tutorial for Beginners in Hindi with MySQL Project. Youtube.
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4. M. O., Jacob Thornton, and Bootstrap. (n.d.). Introduction. Getbootstrap.com.
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